INCH POUND

MIL-C-915/48C
29 September 1989
SUPERSEDING
MIL-C-915/48B
18 June 1985

MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, FOR OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 2SWF

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First	- Copper, conductor, in accordance with ASTM B 286, size 22-7, tin coated. Maximum diameter 0.033 inch.
Second	- Polyethylene insulation, natural, type II, class L, grade 4 in accordance with L-P-390. Nominal thickness 0.010 inch. Colored insulation, one black and one white conductor for each pair.
Third	- Clear polyamide jacket on each conductor in accordance with ASTM D 4066. Nominal thickness 0.003 inch.
Fourth	- Two conductors, one black and one white, cabled together with nominal lay of 2.5 inches to form each pair.
Fifth	- Binder tape over each pair, at manufacturer's option, applied helically with overlap.
Sixth	- Braided shield of AWG no. 34 or no. 36 tin coated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.
Seventh	- Shield insulation of one polyester tape, type G in accordance with MIL-I-631 plus a jacket of clear polyamide in accordance with ASTM D 4066. Nominal thickness of 0.003 inch. Standard identification code applied by method 2.

AMSC N/A FSC 6145

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REQUIREMENTS - Continued

- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.

Eighth - The required number of pairs (see table I) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from center outward. Nonfibrous fillers shall be employed.

Ninth - Synthetic rubber compound filled binder tape applied helically with overlap. (Surface of tape shall be such that cable jacket will adhere to it.)

Tenth - Arctic type polychloroprene jacket (black). (See table I for thickness.) Cable surface marking required.

TABLE I. Details.

Type and size	Number of pairs	Cable jacket thickness minimum (inch)	Overall diameter	
			Minimum (inches)	Maximum (inches)
2SWF-3	3	0.100	0.600	0.625
2SWF-4	4	.090	.600	.625
2SWF-7	7	.110	.780	.815
2SWF-24	24	.125	1.190	1.250

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	17.71
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	500
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	3000
Conductor to shield	1500
Shield to shield	100
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) - leakage 1000 lbf/in², in 6 hours, in³, (test each shipping length)	Zero

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EXAMINATION AND TESTS - Continued

Capacitance	
Mutual capacitance - at 1 MHz, pF/ft, maximum.....	30
Capacitance unbalance - percent, maximum.....	8
Characteristic impedance - at 1 MHz, ohms.....	75 ± 5

	Requirements
Group B:	
Attenuation - at 3 MHz, db/100 feet, maximum.....	3
Cold working (minus 54 ± 2°C).....	No damage
Drip - at 75 ± 1°C.....	Zero
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - lbf/in², minimum.....	1400
Elongation - percent, minimum.....	300
Jacket (cable)	
Tensile strength - lbf/in², minimum.....	1800
Elongation - percent, minimum.....	300
Set - inch, maximum.....	3/8
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - percent of unaged, minimum.	60
Permanence of printing (jacket) - cycles, minimum.....	250
Cable filler removability.....	No failure
Shield-conformance to material coverage and construction.....	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cable aging and compatibility (95 ± 3°C).....	No failure
Cold working (minus 54 ± 2°C).....	No failure

UNIT ORDERING LENGTHS:

All sizes 1000 feet (nominal).

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NOTE: Not for Air Force use.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

Preparing activity:
Navy - SH
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